

Will Doster

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EDUCATION

Massachusetts Institute of Technology

B.S. in Mechanical Engineering, Minor in Computer Science | GPA: 5.0/5.0

Relevant Coursework: Design and Manufacturing, Thermal-Fluids Engineering, Mechanics and Materials, Dynamics and Control, Robotics, Machine Learning, Fundamentals in Programming, Algorithms, Numerical Computation.

Cambridge, MA

Class of 2027

PROFESSIONAL EXPERIENCE

Mecado

Machine Learning Intern

Cambridge, MA

January 2026 – February 2026

- Created a Text-to-CAD environment in NX by programming a plugin for CAD generation and editing in C#.
- Fine-tuned an LLM on semantic descriptions with associated CAD objects represented in a custom Python representation. Performed supervised learning for syntactic understanding and reinforcement learning for geometric reasoning.

Northrop Grumman

Structural and Thermal Engineering Intern

Allegany Ballistics Laboratory, WV

June 2025 – August 2025

- Performed advanced structural and thermal FEA with Abaqus on rocket motor components for a variety of load cases.
- Designed a modular, 3D printed ring igniter housing to reduce manufacturing costs and prevent nozzle clogging.
- Processed material test data and created viscoelastic master curves with MATLAB and Excel for use in FEA analyses.

kW Mission Critical Engineering, Member of WSP

Mechanical Engineering Intern

Atlanta, GA

June 2024 – August 2024

- Designed and produced duct and piping drawings for large cooling systems for data centers with CAD software.
- Performed fluid and thermal load calculations for duct and piping systems to inform design and select equipment.

PROJECTS

MIT Aura Lab: Undergraduate Researcher

Flapping-Wing Robot Design

Cambridge, MA

August 2025 – Present

- Designing and manufacturing a pitching wing aerial-aquatic robot with a differential gearbox to perform multimedia environmental observation tasks. Analyzing weight and lift/thrust creation to enable stable flight and media transitions.
- Programming the flight and motor control system. Creating and executing robot flight and swimming test matrices over a range of wind speeds, flapping frequencies, and pitch angles to quantify efficiency gains from the addition of pitch.

MIT Design and Manufacturing Robotics Competition

Competition Robot Design

Cambridge, MA

January 2025 – May 2025

- Achieved 3rd place out of 150 competitors and received MIT International Design Competition award for robot design.
- Designed, manufactured, and programmed an Arduino for drivable robot with a cascading arm to retrieve object at a height of five feet autonomously and a deployable arm to flip a series of levers at 1 Hz. Presented robot in oral design reviews.

Precision Machine Design Lab: Undergraduate Researcher

Flexure Design

Cambridge, MA

August 2024 – December 2024

- Designed and performed FEA on a machine with linearly variable stiffness via an adjustable boundary condition.
- Integrated concept into flexure design to determine force-displacement curve of artificial muscle for use in robotics.

Vacuum Chuck Design

January 2025 – May 2025

- Designed and manufactured a vacuum workholding system for a mill that met tolerances of $\pm 0.001''$ on machined flexures.

Laser Ablation System Design/Controls

November 2023 – May 2024

- Programmed laser motor controls using LabVIEW & C#, translating mirror angles into a prescribed 3D cutting path.
- CAD designed and manufactured piston system to open protective hood with a maximum force of 5N.

LEADERSHIP AND OTHER EXPERIENCE

Chi Phi Fraternity: *President (Previously: Vice President, Secretary)*

May 2025 – December 2025

- Organized bi-weekly meetings for 40 members. Oversaw all house positions and handled conflict resolution.

MIT Men's Lacrosse Team: *Defense*

August 2023 – Present

- Training 20+ hours weekly for competition on MIT's Division III team. Received Academic All-Conference award.

Global Teaching Labs (Italy): *Teacher*

January 2025 – February 2025

- Prepared and delivered lessons and activities on electricity and magnetism and combinatorics to 10 classes of ~20 students.

Boy Scouts: *Eagle Scout, Senior Patrol Leader*

August 2015 – May 2023

- Designed an outdoor bulletin board for sensory trail for local non-profit. Led group of 20 people during construction.

SKILLS AND INTERESTS

Engineering Skills: CAD (NX, SolidWorks, Fusion 360), FEA (Abaqus), Mill/Lathe/CNC, Metrology, Injection Molding.

Software Skills: MATLAB/Simulink, Arduino, Embedded C/C++, Python, C#, Machine Learning, Excel, LabVIEW, Office.

Interests: Backpacking, Snowboarding, Music, Fitness, Food.